

Question				Marking details	Marks available														
					AO1	AO2	AO3	Total	Maths	Prac									
		(iii)		award (1) for each correct column <table><tr><td>Protons</td><td>Splitting pattern</td><td>Relative peak area</td></tr><tr><td>CH₃CO</td><td>singlet</td><td>3 (or 1)</td></tr><tr><td>(CH₃)₃C</td><td>singlet</td><td>9 (or 3)</td></tr></table>	Protons	Splitting pattern	Relative peak area	CH ₃ CO	singlet	3 (or 1)	(CH ₃) ₃ C	singlet	9 (or 3)		2		2		
Protons	Splitting pattern	Relative peak area																	
CH ₃ CO	singlet	3 (or 1)																	
(CH ₃) ₃ C	singlet	9 (or 3)																	
		(iv)	I	the alcohol has hydrogen bonding (the others do not) (1) therefore more energy is needed to separate the molecules (hence higher boiling temperature) (1)		2		2											
			II	similar – they each show four separate signals (four environments) (1) different – the (four) signals will be in different positions (1) accept any other sensible answers		2		2											
				Question 8 total	5	7	3	15	0	5									